Literature-Implied Partial Subcase for arXiv:1106.5020

FA/Banach run fa_banach_001

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Status

This is a `literature_implied_answer (partial subcase)` packet. It does not claim a new proof of Mazur's general rotation problem. It records that a later theorem of Dilworth and Randrianantoanina [2] excludes a classical family of possible non-Hilbertian counterexamples to the problem as stated by Ferenczi and Rosendal [1].

Source Problem

Ferenczi and Rosendal recall Mazur's rotation problem in the following isomorphic form: if X is a separable Banach space whose isometry group acts transitively on the sphere S_X, must X be Hilbertian, i.e. isomorphic to the separable Hilbert space?

The same source uses the standard hierarchy of symmetry notions. A norm is transitive when every unit vector has full unit-sphere orbit under the surjective linear isometries, and it is almost transitive when every such orbit is dense. Hence

$$\text{transitive} \quad \Longrightarrow \quad \text{almost transitive.}$$

Supporting Literature

Dilworth and Randrianantoanina prove that the spaces ℓ_p, $1 < p < \infty$, $p \neq 2$, and all infinite-dimensional subspaces of their quotient spaces admit no equivalent almost transitive renormings. Their abstract describes this as a step toward the Banach–Mazur rotation problem. The paper's Theorem 2.4 gives the subspace-of-ℓ_p obstruction, and Corollary 2.11 gives the quotient-space formulation.

Theorem 1 (Dilworth–Randrianantoanina)**.** *Let $1 < p < \infty$ and $p \neq 2$. No infinite-dimensional subspace of a quotient space of ℓ_p admits an equivalent almost transitive norm. In particular, ℓ_p itself admits no equivalent almost transitive norm.*

Implication for the Source Problem

Claim 1. *No Banach space isomorphic to ℓ_p, $1 < p < \infty$, $p \neq 2$, nor to an infinite-dimensional subspace of a quotient space of such an ℓ_p, can be a non-Hilbertian counterexample to the isomorphic Mazur rotation problem recorded in arXiv:1106.5020.*

Proof. Suppose that a Banach space Y in one of the listed isomorphic classes had a transitive norm. Choose an isomorphism from the corresponding model space X onto Y and pull the norm back to X. The pulled-back norm is equivalent to the original norm on X, and its isometry group is conjugate to the isometry group of Y with the transitive norm. Thus X would admit an equivalent transitive norm.

Every transitive norm is almost transitive. This would give an equivalent almost transitive renorming of X, contradicting the theorem of Dilworth and Randrianantoanina. Therefore the listed isomorphic classes cannot supply counterexamples to the source problem. $\square$

Verification and Limitations

The verification is a direct literature identification:

- arXiv:1106.5020, page 2, states Mazur's rotation problem for separable Banach spaces with transitive isometry group on the sphere;
- arXiv:1106.5020, page 31, records that transitivity implies almost transitivity;
- arXiv:1310.7139, page 1, states the no-almost-transitive-renorming result in the abstract;
- arXiv:1310.7139, page 5, contains Theorem 2.4 for subspaces of ℓ_p;
- arXiv:1310.7139, page 7, contains Corollary 2.11 for subspaces of quotient spaces of ℓ_p.

This packet does not solve the general Mazur rotation problem. It gives only a classical isomorphic-class exclusion for ℓ_p-based spaces when $1 < p < \infty$ and $p \neq 2$.

Search Bounds

The cheap run indexes were searched for `1106.5020`, `1310.7139`, Dilworth, Randrianantoanina, Mazur rotation, Banach–Mazur rotation, almost transitive renorming, `ell_p`, and subspaces of quotient spaces. No prior packet or attempt for this exact subcase was found.

External arXiv/web search on June 14, 2026 found arXiv:1310.7139 as the relevant later source. No full solution of Mazur's general rotation problem was identified within this bounded check.

Human Review Recommendation

Review as a literature-status partial subcase. In particular, check that the statement is counted only as an isomorphic-class exclusion and that the quotient-space conclusion is restricted to $1 < p < \infty$, $p \neq 2$.

References

[1] V. Ferenczi and C. Rosendal, *On isometry groups and maximal symmetry*, arXiv:1106.5020.

[2] S. J. Dilworth and B. Randrianantoanina, *On an isomorphic Banach–Mazur rotation problem and maximal norms in Banach spaces*, arXiv:1310.7139.